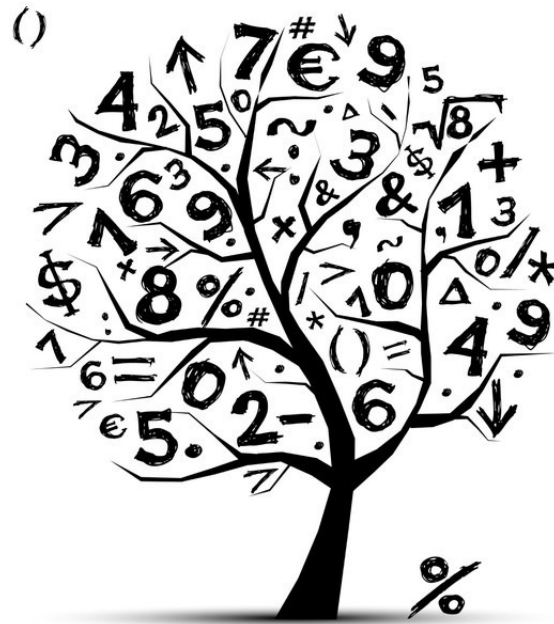


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Products



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1 Definition

Definition 1: product

Given n values $a_j \in \mathbb{R}$ with $j \in \{1, 2, \dots, n\}$, the **product**

$$a_1 a_2 \dots a_n$$

can be indicated in compact form with the product symbol:

$$\prod_{j=1}^n a_j$$

which reads: “product for j from 1 to n of a_j ”. The symbol j is called **product index**.

- The product symbol is therefore very useful when the terms a_j are explicitly defined as a function of the product index j .

Example 1: products

$$\prod_{j=1}^{10} \frac{1}{j} = 1 \frac{1}{2} \frac{1}{3} \frac{1}{4} \frac{1}{5} \frac{1}{6} \frac{1}{7} \frac{1}{8} \frac{1}{9} \frac{1}{10}$$

$$\prod_{j=3}^n j^2 = 3^2 4^2 5^2 \dots n^2$$

2 Properties

Observation 1: product with constant term

For every natural number $n \geq 1$ and real number c , we have:

$$\prod_{j=1}^n c = c^n \quad (1)$$

Proof. We have:

$$\underbrace{c \ c \ \dots \ c}_{c \text{ multiplied } n \text{ times}} = c^n$$

□

Observation 2: product by a constant

For every product $\prod_{j=1}^n a_j$ and real number c , we have:

$$\prod_{j=1}^n (c a_j) = c^n \prod_{j=1}^n a_j \quad (2)$$

Proof. From the commutative and associative properties of the product, we have:

$$\begin{aligned} \underbrace{c a_1 c a_2 \dots c a_n}_{=\prod_{j=1}^n (c a_j)} &= \underbrace{\underbrace{(c c \dots c)}_{=\prod_{j=1}^n c, \text{ } c \text{ multiplied } n \text{ times}} (a_1 a_2 \dots a_n)}_{=c^n \prod_{j=1}^n a_j} \end{aligned}$$

□

Observation 3: product of products

For every pair of products $\prod_{j=1}^n a_j$ and $\prod_{j=1}^n b_j$, we have:

$$\prod_{j=1}^n a_j \prod_{j=1}^n b_j = \prod_{j=1}^n (a_j b_j) \quad (3)$$

Proof. We have:

$$\underbrace{a_1 a_2 \dots a_n b_1 b_2 \dots b_n}_{=\prod_{j=1}^n a_j \prod_{j=1}^n b_j} = \underbrace{a_1 b_1 a_2 b_2 \dots a_n b_n}_{=\prod_{j=1}^n (a_j b_j)}$$

□

Observation 4: decomposition

For every product $\prod_{j=1}^{n+m} a_j$, we have:

$$\prod_{j=1}^{n+m} a_j = \prod_{j=1}^n a_j \prod_{j=n+1}^{n+m} a_j \quad (4)$$

Observation 5: index translation

For every product $\prod_{j=1}^n a_j$ and natural number $m \geq 1$, we have:

$$\prod_{j=1}^n a_j = \prod_{j=1+m}^{n+m} a_{j-m} = \prod_{j=1-m}^{n-m} a_{j+m} \quad (5)$$

Observation 6: index reflexion

For every product $\prod_{j=1}^n a_j$, we have:

$$\prod_{j=1}^n a_j = \prod_{j=1}^n a_{n-j+1} = \prod_{j=0}^{n-1} a_{n-j} \quad (6)$$

Proof. The three properties are simply different notations and/or arrangements of the terms in the products. $\square$

3 Factorial

Definition 2: Factorial

The **factorial** of a natural number n , denoted as $n!$, is the product of the first n positive natural numbers. By convention, the factorial of 0 is defined as 1.

- Formally, the factorial is defined as follows:

$$n! = \begin{cases} 1 & \text{if } n = 0, \\ \prod_{j=1}^n j & \text{if } n \geq 1. \end{cases} \quad (7)$$

- A closed formula for the factorial is not known but Stirling's approximation provides an asymptotic approximation.

Observation 7

For every pair of natural numbers $n \geq 0$ and $k \in \{0, 1, \dots, n\}$, we have:

$$\frac{n!}{(n-k)!} = \prod_{j=1}^k (n-j+1) = \prod_{j=n-k+1}^n j \quad (8)$$

- Formula (8) is the product of k factors, from n and decreasing by one unit at a time.
- By convention, a product with no factors (i.e., whose upper index is smaller than its lower index, as $\prod_{j=1}^0 a_j$) is equal to 1. This convention is consistent with $0! = 1$ and makes formula (8) valid also for $k = 0$ and $k = n$.

Proof. We have:

$$\frac{n!}{(n-k)!} = \frac{\prod_{j=1}^n j}{\prod_{j=1}^{n-k} j} = \frac{\prod_{j=1}^{n-k} j \prod_{j=n-k+1}^n j}{\prod_{j=1}^{n-k} j} = \prod_{j=n-k+1}^n j = \prod_{j=1}^k (n-j+1)$$

□

Example 2: use of the observation

$$\frac{100!}{98!} = \frac{100!}{(100-2)!} = \prod_{j=1}^2 (100-j+1) = 100 \cdot 99 = 9.900$$